

1. A single phase 242V source supplies power through a 250ft length of underground power cable to a constant impedance load. Each conductor of the power cable has a distributed series impedance of $0.00125\angle 60^\circ \Omega/\text{foot}$, and there is a distributed shunt admittance of $0.000003\angle 85^\circ \text{S}/\text{foot}$ between the two power cable conductors. The load has an equivalent impedance of $29.754\angle 58^\circ \Omega$. Compute the following quantities using the series impedance model for the power cable.
 - a. Source current. (7.966A)
 - b. Complex power supplied by the source. ($1927.803\angle 58.041^\circ \text{VA}$)
 - c. Source power factor. (0.529310 lagging)
 - d. Real and reactive power supplied by the source. (1020.406W, 1635.603VAR)
 - e. Load voltage. (237.024V)
 - f. Complex power consumed by the load. ($1888.164\angle 58^\circ \text{VA}$)
 - g. Real and reactive power consumed by the load. (1000.575W, 1606.135VAR)
 - h. Load power factor. (0.529919 lagging)
 - i. Real power loss in the power cable. (19.831W)
 - j. Power system efficiency. (98.057%)

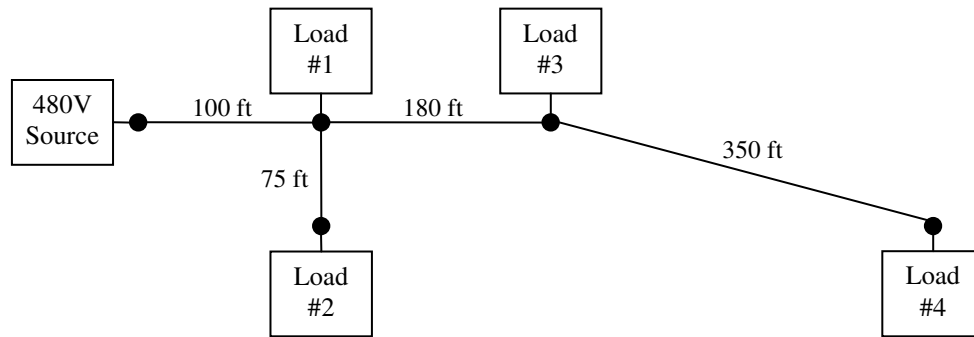
2. Repeat the previous problem using the pi-model for the power cable instead of the series impedance model.
 - a. Source current. (7.825A)
 - b. Complex power supplied by the source. ($1893.613\angle 57.257^\circ \text{VA}$)
 - c. Source power factor. (0.540865 lagging)
 - d. Real and reactive power supplied by the source. (1024.189W, 1592.736VAR)
 - e. Load voltage. (237.069V)
 - f. Complex power consumed by the load. ($1888.874\angle 58^\circ \text{VA}$)
 - g. Real and reactive power consumed by the load. (1000.951W, 1601.856VAR)
 - h. Load power factor. (0.529919 lagging)
 - i. Real power loss in the power cable. (23.238W)
 - j. Power system efficiency. (97.731%)

3. A three phase 8KV source supplies power to two parallel loads through a length of underground power cable. The type of power cable used has four conductors with each conductor having a total series impedance of $6.3725\angle 76.318^\circ \Omega$. The first load is Y-connected with per phase impedance $50\angle -37^\circ \Omega$, and the second load is Δ -connected with per phase impedance $245\angle 73^\circ \Omega$. Compute the following quantities.
 - a. Source power factor. (0.994897 lagging)
 - b. Real and reactive power supplied by the source. (1203.5043KW, 122.0507KVAR)
 - c. Load voltage. (7.7323KV)
 - d. Complex power absorbed by each of the two loads. ($1195.78\angle -37^\circ \text{KVA}$, $732.1102\angle 73^\circ \text{KVA}$)
 - e. Power system efficiency. (97.136%)

4. A single phase generator supplies power through a 285ft length of underground power cable to a constant impedance load. The generator is rated 7.5KVA, 240V and has 18.59% stator winding synchronous reactance with an X/R ratio of 118. Observations show that the generator operates with an output voltage of 235.25V and supplies 4.125KW and 2.630KVAR to the system. The power cable has a distributed series impedance of $1.2047\angle 10^\circ \Omega/1000\text{ft}$. Compute the following quantities.
 - a. Source current. (20.795A)
 - b. Generator no-load voltage required to operate the generator at the given conditions. (252.6533V)
 - c. Load voltage. (228.671V)
 - d. Equivalent impedance the load. ($10.9963\angle 33.206^\circ \Omega$)
 - e. Real and reactive power consumed by the load. (3.9788KW, 2.6042KVAR)
 - f. Load power factor. (0.836709 lagging)
 - g. Real power loss in the power cable. (146.219W)
 - h. Power system efficiency. (96.455%)

5. A single phase 350V source supplies 9.2282KVA at a 0.98757 leading power factor through a 500 meter length of underground power cable to a constant impedance load. Each conductor of the power cable has a distributed series impedance of $0.1936\angle 22^\circ\Omega/\text{mile}$. Compute the following quantities.
- Source current. (26.366A)
 - Complex power supplied by the source. ($9.2282\angle -9.043^\circ\text{KVA}$)
 - Real and reactive power supplied by the source. (9.1135KW, -1.4505KVAR)
 - Load voltage. (347.286V)
 - Equivalent impedance the load. ($13.17161\angle -9.313^\circ\Omega$)
 - Complex power consumed by the load. ($9.1567\angle -9.313^\circ\text{KVA}$)
 - Real and reactive power consumed by the load. (9.0360KW, -1.4818VAR)
 - Load power factor. (0.986819 leading)
 - Real power loss in the power cable. (77.539W)
 - Power system efficiency. (99.149%)
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6. A three phase Y-connected source supplies power to a Δ -connected constant impedance load through a 800 foot length of power cable. The type of power cable used has three conductors with each conductor having a distributed series impedance of $1.5\angle 70^\circ\Omega/1000\text{ft}$. If the source operates with $V_{CA}=965\angle -20^\circ\text{V}$ and $I_B=34\angle 80^\circ\text{A}$, compute the following quantities.
- Source power factor. (0.984808 leading)
 - Real and reactive power supplied by the source. (55.9652KW, -9.8682KVAR)
 - Load voltage. (955.267V)
 - Per phase impedance of the Δ -connected load. ($48.6639\angle -14.178^\circ\Omega$)
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7. The beacon lamp of a lighthouse is supplied by a diesel engine powered single phase generator located in a small building at the base of the lighthouse. The generator is rated 5KVA, 485V and has 4.5% stator winding synchronous reactance with 250 X/R ratio. A 400ft length of power cable is used to connect the beacon lamp to the generator. Each conductor of this power cable has a distributed series impedance of $0.0033\angle 25^\circ\Omega/\text{ft}$, and the power cable has a distributed shunt admittance of $0.00002\angle 87^\circ\text{S}/\text{foot}$ between the two conductors. The beacon lamp can be modeled by an impedance of value $70\angle 70^\circ\Omega$, and requires a minimum voltage of 475V to maintain the desired light output. Compute the following quantities using the series impedance model for the power cable.
- Generator no-load voltage required to operate the beacon lamp at its required voltage. (501.247V)
 - Real and reactive power supplied by the generator. (1.2126KW and 3.0802KVAR)
 - Real and reactive power consumed by the beacon lamp. (1.1024KW and 3.0288KVAR)
 - Power system efficiency. (90.914%)
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8. Repeat the previous problem using the pi-model for the power cable instead of the series impedance model.
- Generator no-load voltage required to operate the beacon lamp at its required voltage. (491.195V)
 - Real and reactive power supplied by the generator. (1.2611KW and 1.2136KVAR)
 - Real and reactive power consumed by the beacon lamp. (1.1024KW and 3.0288KVAR)
 - Power system efficiency. (87.418%)
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9. A single phase 480V source supplies power to a remotely located power distribution panel through a 2350ft length of underground power cable. The power cable has a distributed series impedance of $0.104\angle 15^\circ\Omega/1000\text{ft}$. Three loads are connected at the power distribution panel. The 1st load has an equivalent impedance of $10.15\angle -85^\circ\Omega$, the 2nd load has an equivalent impedance of $8.75\angle 75^\circ\Omega$, and the 3rd load has an equivalent impedance of $30.25\angle 18^\circ\Omega$. Compute the following quantities.
- Source current. (34.491A)
 - Real and reactive power supplied by the source. (15.7604KW, 5.0700KVAR)
 - Source power factor. (0.951955 lagging)

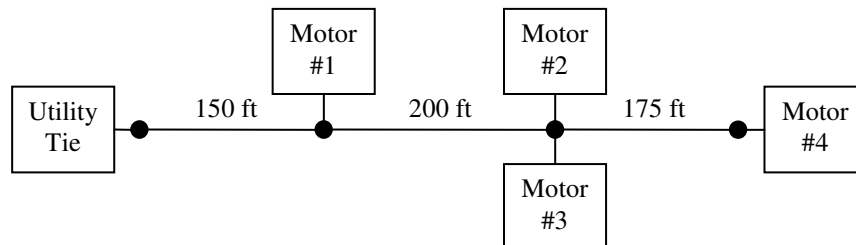
- d. Distribution panel voltage. (471.581V)
 - e. Real and reactive power consumed by the 1st load. (1.9096KW, -21.8268KVAR)
 - f. Real and reactive power consumed by the 2nd load. (6.5781KW, 24.5498KVAR)
 - g. Real and reactive power consumed by the 3rd load. (6.9919KW, 2.2718KVAR)
 - h. Real power loss in the power cable. (280.844W)
 - i. Power system efficiency. (98.218%)
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10. A single phase 280V emergency power source located in the basement of a hospital is used to supply backup power to life support equipment in the intensive care wards located on the 13th and 16th floors of the building. A 1400ft length of power cable is used to connect the power source to the 13th floor equipment, and a 200ft length of power cable is run from the 13th floor equipment to the 16th floor equipment. Each conductor of the type of power cable used has a distributed series impedance of $0.001\angle 25^\circ\Omega/\text{foot}$. The life support equipment on both the 13th and 16th floors is identical and can be modeled by impedance of value $170\angle 60^\circ\Omega$. Compute the following quantities.
- a. Source current. (3.204A)
 - b. Apparent power supplied by the source. (897.114VA)
 - c. Source power factor. (0.516392 lagging)
 - d. Voltage at the 13th floor equipment. (272.601V)
 - e. Voltage at the 16th floor equipment. (272.076V)
 - f. Real and reactive power consumed by 13th floor equipment. (218.562W, 378.561VAR)
 - g. Real and reactive power consumed by 16th floor equipment. (217.722W, 377.105VAR)
 - h. Real power loss in the power cable. (26.979W)
 - i. Power system efficiency. (94.176%)
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11. A single phase power system for a small manufacturing facility is supplied by a 240V tie from the electric utility company. Inside the manufacturing facility, there are two power distribution panels to which all of the building's loads are connected. The 1st distribution panel is connected to the electric utility company tie through a 500ft power cable, and a 350ft power cable connects the two distribution panels together. The loads connected to the 1st distribution panel have a net equivalent impedance of $300\angle 50^\circ\Omega$, and the loads connected to the 2nd distribution panel have a net equivalent impedance of $20\angle 75^\circ\Omega$. Each conductor of the type of power cable used in the building has an impedance of $0.0003\angle 20^\circ\Omega/\text{ft}$. The electric utility company tie has a source impedance of $0.15\angle 85^\circ\Omega$. Compute the following quantities.
- a. Real and reactive power supplied by the electric utility company to the manufacturing facility. (0.9016KW, 2.8235KVAR)
 - b. Current in each of the two power cables. (12.444A, 11.725A)
 - c. Voltage at each of the two distribution panels. (235.914V, 234.493V)
 - d. Total real power loss in the system power cabling. (70.785W)
 - e. Power system efficiency. (92.149%)
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12. The manufacturing facility in the previous problem is being expanded by adding a 3rd power distribution panel that is connected to the 1st power distribution panel through a 200ft power cable of the same type as the other power cables. The net load connected to this 3rd distribution panel has an equivalent impedance of $50\angle 45^\circ\Omega$. Compute the following quantities.
- a. Real and reactive power supplied by the electric utility company to the manufacturing facility. (1.6973KW, 3.5666KVAR)
 - b. Voltage at each of the three distribution panels. (234.136V, 232.725V, 233.627V)
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13. The block diagram for a commercial single phase power system is given below, in which a 480V source supplies power to four remotely located loads through a network of underground power cables with lengths shown. The type of power cable used has a distributed series impedance of $0.1673\angle 61.442^\circ\Omega/1000\text{ft}$. All four of the loads are identical with equivalent impedance $15\angle 63^\circ\Omega$.



Compute the following quantities.

- Real and reactive power supplied by the source. (27.6928KW, 54.3226KVAR)
- Source power factor. (0.454174 lagging)
- Voltage at each of the four loads. (477.876V, 477.476V, 475.969V, 474.119V)
- Current in each of the four power cables. (127.029A in the 100ft cable, 31.832A in the 75ft cable, 63.339A in the 180ft cable, 31.608A in the 350ft cable)
- Real and reactive power flow on both ends of the 350ft power cable, assuming a Load #3 to Load #4 power flow direction. (6.8314KW, 13.4039KVAR, 6.8035KW, 13.3525KVAR)
- Efficiency of the 350ft power cable. (99.591%)
- Power system efficiency. (99.202%)

14. In a manufacturing facility, parts are moved on a series of conveyer belts from one end of the building to the other end of the building for the assembly and packaging workers. These conveyer belts are driven by 4 different single phase motors. As shown in the block diagram below, the motors are powered from an electric utility company tie and are connected to this source using a network of power cables. Each conductor of the type of power cables used has a distributed series impedance of $0.3\angle 40^\circ \Omega/1000\text{ft}$. The motors are identical and each can be modeled by a constant impedance of value $50\angle 70^\circ \Omega$. The electric utility company tie is rated 240V with a short circuit current of 3400A and an X/R of 10.

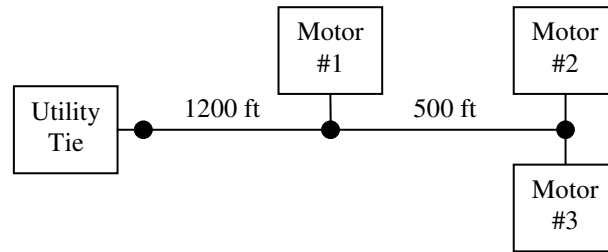


Compute the following quantities.

- Voltage, current, and power factor at the electric utility tie point. (238.710V, 18.882A, 0.3481 lagging)
- Real and reactive power supplied by the electric utility company to the system. (1.5690KW, 4.2254KVAR)
- Voltage at each of the four motors. (237.235V, 235.764V, 235.764V, 235.336V)
- Current in each of the three power cables. (18.882A in the 150ft cable, 14.137A in the 200ft cable, 4.707A in the 175ft cable)
- Real and reactive power flow on both ends of the 200ft power cable, assuming a Motor #1 to Motor #4 power flow direction. (1.1594KW, 3.1471KVAR, 1.1411KW, 3.1316KVAR)
- Power system efficiency. (97.149%)

15. In a manufacturing facility, parts are moved on a series of conveyer belts from one end of the building to the other end of the building for the assembly and packaging workers. The conveyer belts are driven by 3 different three phase motors. The motors are powered by a three phase electric utility company tie, and they are connected to this source using a network of power cables. The block diagram below shows the source, motors, and cables of this three phase power system. The three motors are identical and each can be

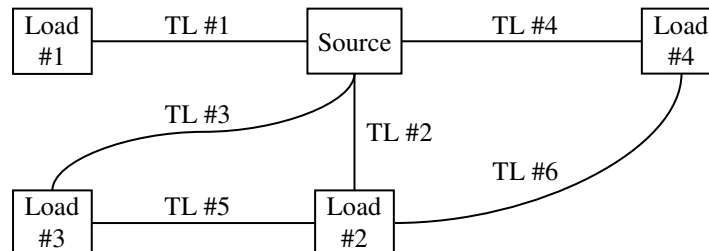
modeled by a Y-connected load with per phase impedance $10\angle 65^\circ\Omega$. The type of power cable used has three solid aluminum wires enclosed and separated by a special type of polyvinyl insulating material. Each aluminum wire has an equivalent distributed series impedance of $0.08133\angle 10.37^\circ\Omega/1000\text{ft}$. The electric utility company tie is rated 208V with a short circuit current of 2600A and an X/R of 12.



Compute the following quantities.

- Voltage, current, and power factor at the electric utility company tie point. (205.408V, 34.864A, 0.447574 lagging)
- Real and reactive power supplied by the electric utility company to the system. (5.5516KW, 11.0920KVAR)
- Voltage at each of the three motors. (201.920V, 200.969V, 200.969V)
- Current in each of the two power cables. (34.864A, 23.206A)
- Total real power loss in the system power cables. (414.686W)
- Efficiency of the 1200ft power cable. (93.694%)
- Power system efficiency. (92.530%)

16. The block diagram for a single phase power system is given below in which a 556V source supplies power to four loads using a network of six transmission lines.



Load #1 has an equivalent impedance $70\angle 50^\circ\Omega$. Load #2 has an equivalent impedance $30\angle 60^\circ\Omega$. Load #3 has an equivalent impedance $80\angle 30^\circ\Omega$. Load #4 has an equivalent impedance $90\angle -70^\circ\Omega$. The transmission lines all use the same design having a distributed series impedance of $1\angle 20^\circ\Omega/1000\text{ft}$, but have the following different lengths in consecutive order: 730ft, 550ft, 1050ft, 380ft, 250ft, and 950ft.

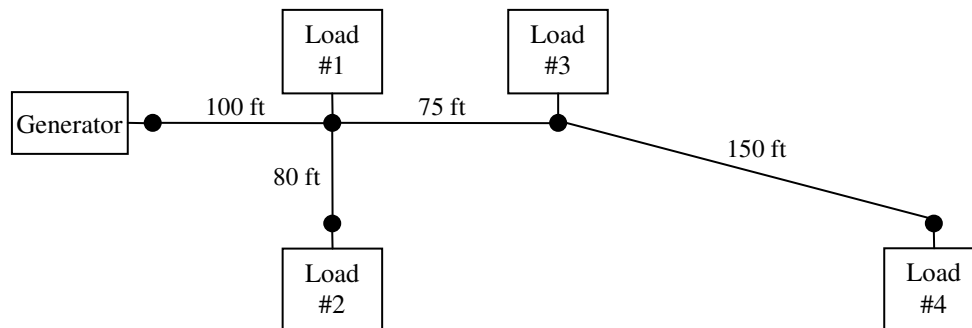
An analysis of the equivalent circuit for this system was done for you by another engineering student, and the load voltage phasors calculated in the equivalent circuit are listed below.

$$\begin{aligned}\vec{V}_{L1} &= 551.0045\angle 0.296^\circ\Omega, & \vec{V}_{L2} &= 550.1309\angle 0.339^\circ\Omega \\ \vec{V}_{L3} &= 549.8889\angle 0.299^\circ\Omega, & \vec{V}_{L4} &= 554.3103\angle -0.077^\circ\Omega\end{aligned}$$

Assume that the source voltage phasor in the equivalent circuit had a 0° phase angle in the engineering student's analysis work. Compute the following quantities.

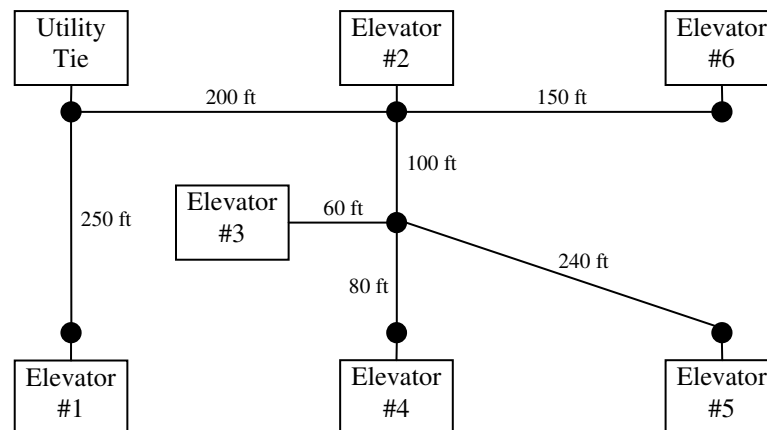
- Real and reactive power supplied by the source to the system. (12.5197KW, 10.8227KVAR)
- Power system efficiency. (98.030%)
- Real and reactive power flow on both ends of TL#5, assuming the direction is from Load #2 to Load #3. (789.617W, -611.588VAR on the Load #2 end, and 788.842W, -611.870VAR on the Load #3 end)
- Efficiency of TL#3. (98.381%)

17. The block diagram for a three phase power system is given below. In this system, a Y-connected generator supplies power to four remotely located loads through a network of underground power cables. The generator is rated 100KVA, 485V and has 15.25% stator winding synchronous reactance with X/R of 122. The type of power cable used has three conductors with each conductor having a distributed series impedance of $0.00164\angle 25^\circ\Omega/\text{meter}$. All four of the loads are identical, being Y-connected with a per phase impedance of $15\angle 65^\circ\Omega$. Assume the generator operates with a no-load voltage of 525.85V.



Compute the following quantities.

- Real and reactive power supplied by the generator. (26.6862KW, 55.5888KVAR)
 - Voltage at each of the four loads. (479.235V, 478.258V, 477.405V, 475.582V)
 - Current in each of the four power cables. (73.534A in the 100ft cable, 18.408A in the 80ft cable, 36.680A in the 75ft cable, 18.305A in the 150ft cable)
 - Real and reactive power flow on both ends of the 75ft power cable, assuming a Load #1 to Load #3 power flow direction. (12.9994KW, 27.5324KVAR, 12.8622KW, 27.4685KVAR)
 - Power system efficiency. (96.338%)
18. The block diagram drawn below depicts the three phase power system used to operate the elevators of a high rise office building. The power system consists of a 480V electric utility company tie that supplies power to six elevators through a network of power cables.

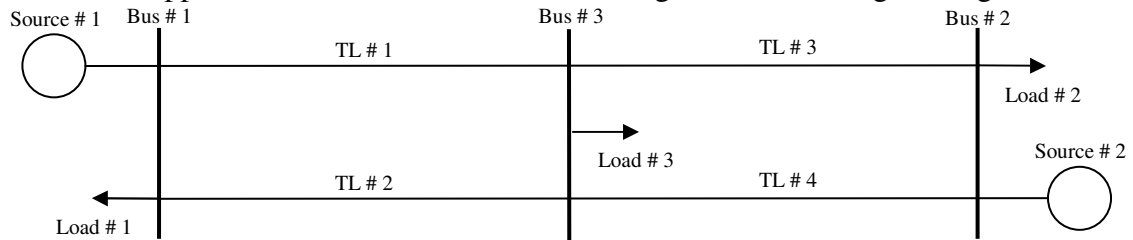


The type of power cable used in this system has three conductors, with each conductor having a distributed series impedance of $0.448\angle 15^\circ\Omega/1000\text{ft}$. Each elevator can be modeled as a Y-connected load with per phase impedance $25\angle 75^\circ\Omega$. The utility tie is rated 480V with a short circuit MVA of 100 and X/R of 7.5.

Assuming all elevators are running at full speed and full load, compute the following quantities.

- Real and reactive power supplied by the utility tie. (15.0513KW, 52.5688KVAR)
- Voltage at each of the six elevators. (478.664V, 475.403V, 473.866V, 473.781V, 473.100V, 474.764V)
- Real and reactive power on both ends of the 200ft power cable, assuming a Utility Tie to Elevator #2 power flow direction. (12.6396KW, 43.7057KVAR, 11.8612KW, 43.4971KVAR)
- Power system efficiency. (93.089%)

19. The one line diagram drawn below shows part of an electric utility three phase power system in which three different loads are supplied from two different sources using a network of high voltage transmission lines.

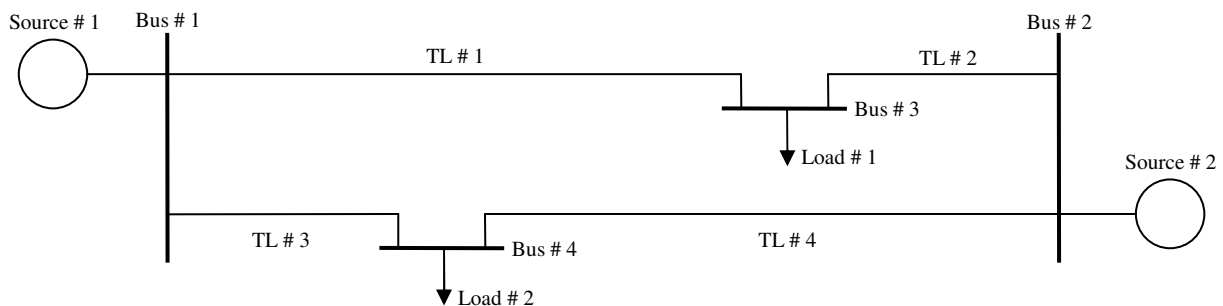


Source #1 operates at 115.259KV, Source #2 operates at 115.385KV, and the voltage at Source #1 lags the voltage at Source #2 by 1.2° . Both Load #1 and Load #2 are identical Y-connected loads with per phase impedance $332.87\angle 70^\circ \Omega$, and Load #3 is a Δ -connected load with per phase impedance $572.48\angle 39^\circ \Omega$. All four transmission lines are the same design having a distributed per phase series impedance of $0.664687\angle 85.357^\circ \Omega/\text{mile}$, but have different lengths. The transmission line lengths in consecutive order are 30 miles, 40 miles, 25 miles, 50 miles. Assume that the transmission lines in this system can each be modeled by a single lumped series impedance.

Compute the following quantities.

- Voltage at Load #3. (112.9959KV)
- Real and reactive power supplied by each source. (27.183122MW, 59.430757MVAR, 52.325132MW, 59.986237MVAR)
- Current in TL #2 and the real and reactive power flow on both ends of TL #2, assuming direction is towards bus #3. (55.319A, 5.799989MW, 9.397801MVAR, 5.780231MW, 9.154518MVAR)
- Current in TL #3 and the real and reactive power flow on both ends of TL #3, assuming direction is towards bus #3. (149.006A, 25.763639MW, 14.934428MVAR, 25.674044MW, 13.831220MVAR)
- Power system efficiency. (99.773%)

20. The one line diagram below shows part of an electric utility three phase power system in which two loads are supplied from two sources using a simple loop network of four high voltage transmission lines.



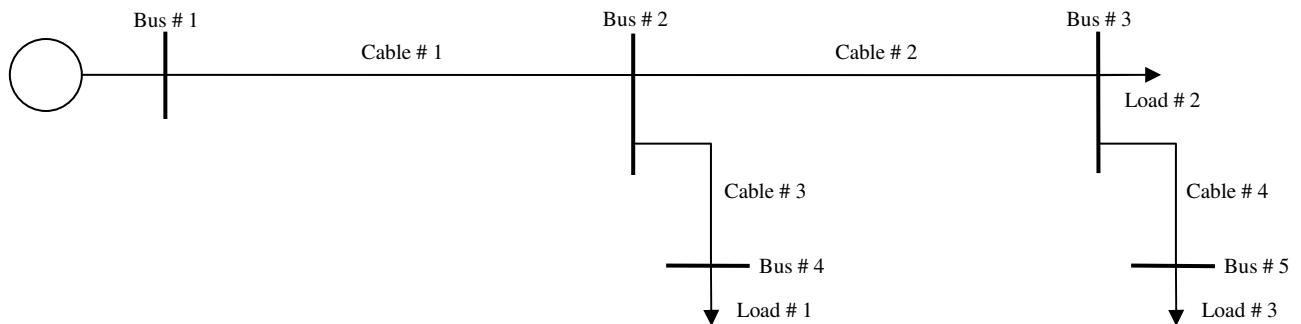
Transmission line TL#1 has a total per phase series impedance of $35\angle 85^\circ \Omega$, and transmission line TL#2 has an impedance that is 0.25 times that of TL#1. Transmission line TL#4 has a total per phase series impedance of $29.5\angle 87^\circ \Omega$, and transmission line TL#3 has an impedance that is 0.35 times that of TL#4. Load #1 is Y-connected with per phase impedance $350\angle 67^\circ \Omega$, and Load #2 is Y-connected with per phase impedance $410\angle 47^\circ \Omega$. Measurements taken in this system show that Source #1 operates at 140.575KV, Source #2 operates at 141.013KV, and that the voltage at Source #1 leads the voltage at Source #2 by 0.85° .

Compute the following quantities.

- Real and reactive power supplied by each source. (41.646270MW, 32.623067MVAR, 11.856830MW, 54.167066MVAR)
- Voltage at each of the two loads. (138.2899KV, 138.6938KV)
- Current in each of the four transmission lines. (55.888A, 181.942A, 161.387A, 45.447A)

- d. Real and reactive power flow on both ends of TL #1, assuming direction is towards bus #3. (10.734007MW, 8.363615MVAR, 10.705424MW, 8.036904MVAR)
- e. Real and reactive power flow on both ends of TL #2, assuming direction is towards bus #3. (10.719967MW, 43.125378MVAR, 10.644233MW, 42.259736MVAR)
- f. Real and reactive power flow on both ends of TL #3, assuming direction is towards bus #4. (30.912262MW, 24.259452MVAR, 30.870040MW, 23.453794MVAR)
- g. Real and reactive power flow on both ends of TL #4, assuming direction is towards bus #4. (1.136863MW, 11.041688MVAR, 1.127296MW, 10.859148MVAR)
- h. Real power loss in each of the four transmission lines. (28.5835KW, 75.7339KW, 42.2227KW, 9.5665KW)
- i. Power system efficiency. (99.708%)

21. The one line diagram below shows a small commercial single phase power system in which three loads are supplied from a 350V source through a network of power cables.



The type of power cable used has a distributed series impedance of $0.000628\angle 30^\circ \Omega/\text{foot}$. The lengths of the power cables in consecutive order are 500ft, 400ft, 300ft, and 200ft. Load #1 and Load #2 are identical and can be modeled by a constant impedance of value $50\angle 70^\circ \Omega$. Load #3 can be modeled by a constant impedance of value $90\angle 60^\circ \Omega$. Compute the following quantities.

- a. Real and reactive power supplied by the source. (2.3907KW, 5.6495KVAR)
- b. Voltage at each of the three loads. (344.628V, 343.469V, 343.054V)
- c. Current in each of the four power cables. (17.527A in Cable #1, 10.643A in Cable #2, 6.893A in Cable #3, 3.812A in Cable #4)
- d. Power system efficiency. (95.085%)